

# Curriculum Vitae Zemzem Firas

## Personal data

**Birth** : 13/10/1998
**Nationality** : Tunisian
**Phone** : +216 94 95 77 39 - +216 22 91 55 87
**E-mail** : zemzemfiras@gmail.com
firas.zemzem@isbm.u-monastir.tn
**Linkedin** : https://www.linkedin.com/in/firaszemzem/
**Researchgate** : researchgate.net/profile/Firas-Zemzem
**ORCID** : https://orcid.org/0009-0004-7791-6364
**Portfolio** : https://zemzemfiras1.github.io/Portfolio/
**Langues** : Arabic, French, English
**Note** : Limb-girdle muscular dystrophy type 2A (LGMD2A)

## Diploma

**Currently : PhD Candidate,**
Laboratory of Cytogenetics, Molecular Genetics and Biology of Reproduction CHU Farhat Hached Sousse,
Higher institute of Biotechnology of Monastir,
University of Monastir, Tunisia

**2020-2022 : Master Degree in Genetic, Biodiversity & Valorisation**
Higher Institute of Biotechnology of Monastir, Tunisia

**2017-2020 : Bachelor's Degree in fundamental life science : Molecular and Cellular Biology**
Higher Institute of Biotechnology of Monastir, Tunisia

**2017 : Post High School Technical Degree (Experimental Sciences)**

## Scientific research

**Zemzem, F.**, Chouk, H., Hamadou, WS. et al. Socio-genomics of intellectual disability in African populations: top research-producing countries. J Community Genet 17, 130 (2026). https://doi.org/10.1007/s12687-026-00966-9

Igiraneza, C., de Castro Silva, D., **Zemzem, F.** et al. Workflow management systems and reproducible pipelines for Plasmodium falciparum genomic surveillance: tools, design principles and infrastructure challenges. Malar J (2026). https://doi.org/10.1186/s12936-026-06123-4

Coetzer, K.C., **Zemzem, F.**, Akurut, E. *et al.* RareInsight simplifies the communication of genetic results for rare disease patients. *Sci Rep* 15, 24442 (2025).

https://doi.org/10.1038/s41598-025-09744-y
Foddha H, Seo GH, Lee H, **Zemzem F**, Khelil AH, et al. (2020) Exome sequencing reveals a homozygous frameshift variant in CAPN3 in a Tunisian patient with a neuromuscular disorder. Ann Mol Genet Med 6(1): 001-004. DOI:https://dx.doi.org/10.17352/amgm.00001

## Professional experience

**2024 : Awarded Second place in African Society for Bioinformatics and Computational Biology Omics Codeathon April 2024** Developing an open-source, interactive Shiny dashboard tailored to both clinicians and patients aiming to facilitate Data finding and reporting.

https://github.com/omicscodeathon/rareinsight

**2023 : Participation at ASBCB Omics Codeathon – October 2023**
Prediction of novel antimicrobial resistant genes in Acetinobacter baumannii using Machine Learning, Homology Modelling and Molecular Docking

https://www.asbcb.org/event/codeathon/asbcb-omics-codeathon-october-2023

**2022 : Master's subject : Analysis of the effect of mutations on protein's stability using bioinformatic tools**

Predicting the effect of mutations using classic predictive tools
Simulating native and mutant proteins by molecular dynamics using GROMACS

## Competences

- Redaction of reports
- Bibliography for scientific reports
- Use of : Windows, Linux
- Data Bank used : NCBI, GenBank, Refseq, OMIM, Uniprot, PDB, STRING, PhosphoSitePlus ...
- Software used : GROMACS, PYMOL, VMD, USCF CHIMERA, BIOVIA DSV, AutoDock Vina, Avogadro ...
- Software used : Microsoft word, Microsoft excel, Microsoft powerpoint, LibreOffice Writer
- Language : Python, R, Bash
- Nextlow
- NGS Analysis

## Training & Workshops

**2025** : « Ensembl Train the Trainer workshop »
« Genomics for Educators - Wellcome connecting science »
« Train the Trainer- Design Genomics and bioinformatics Training - Wellcome connecting scien»
« Gene Therapy, Towards a new era of medicine»
« Policy Advocacy in Genetics and Genomics- Pan-African

PGS Education and Research Initiative PAPERI »
« Genomics our way - an introduction to Genomics Research with indigenous australians online course» « Genetic Counselling workshop - Human Genome Meeting »

**2024 :** « Advancing north African bioeconomy through biodiversity genomics - African BioGenome Project »
« In-Silico Drug Design and Repurposing for Rare Diseases workshop - IPT Tunisia »
« Data sharing and databases Workshop-TSHG »
« Attendance in the 1st Congress of the Tunisian Society of Human Genomics-TSHG »
« The Next Generation Sequencing Competence Network Summer School »
« the eLwazi - Open Data Science Platform (eLwazi-ODSP) Research Data Management Course »
« Accepted in arabs Neuroscience Course »
« Interpreting Genomics Variation course-future Learn »
« Fundamentals of Data Science in Precision medicine and cloud computing- Stanford Medicine »
« Being a Nextflow Ambassador »
« Genomics science and data careers with wellcome sanger institute »
« *I*ntroduction to functional Genomics -Jackson Lab » « 36th Course on Clinical Genomics and NGS organized by ESHG »
« Second place in African Society for Bioinformatics and Computational Biology Omics Codeathon April 2024 » « Participation in the Malaysia-Japan Bilateral workshop on BioInformatics and Bioressourrces. »
« Analysis and interpretation of ngs data in Clinics 4 weeks training »
« The era of genetically modified humans (gene therapy and its applications)workshop»
« Attended the International Parkinson’s Conference (Parkinson’s-2024 »
« Member of Tunisian Society of Human Genomics - TSHG »
« Member of DINS-SOCIO GENOMICS Federated Research Project »
« 2nd Saudi Rare Disease Summit »
« Seminar "Biotools Nanopore Day »

**2023 :** « Python Essentials 1 Course Cisco Networking Academy »
« 2023 Genome Sequencing Bioinformatics Course | H3ABioNet | Wellcome connecting Science »
« Bioinformatics For Biologists : An Introduction To Linux, Bash Scripting, And R »
« Python For DataScience | Great Learning »
« Bioinformatics For Biologists : Analysing And Interpreting Genomics Datasets | Wellcome Connecting Science »
« A practical Guide For Sars-Cov-2 Whole Genome

Sequencing | Wellcome Connecting Science And Cog-Train

»

**2022 :** « Pitching d’un projet »

**2021 :** « In Silico Molecular Docking »

« Next Generation Sequencing »

**2020 :** « Phylogénie Workshop »

**2019 :** «Debate competition « OGM pour ou contre »

« Du BMC au Pitching »

« Practical molecular graphics and visualization »
« Practical comparative structural modeling »
Training in The laboratory LR11ES41 : Génétique, Biodiversité et Valorisation des Bioressources